

Report - Some local definability theory for holomorphic functions

Quentin Moreau

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1 Introduction

The study of locally definable holomorphic functions lies at the interface between model theory and analytic geometry. A major development in this direction was the emergence of $\mathcal{o}$ -minimality, which provides a framework in which definable sets admit strong geometric regularity properties. In particular, in $\mathcal{o}$ -minimal expansions of the real field, definable sets have finitely many connected components and admit cell decomposition (see van den Dries [1], van den Dries and Miller [2]).

Important examples arise from expansions of the real field by restricted analytic functions and the exponential function. Wilkie proved that such structures are model complete and $\mathcal{o}$ -minimal [3], thereby showing that analytic functions can be incorporated into first-order structures while preserving geometric tameness. These results motivate the investigation of definability phenomena in analytic settings.

Holomorphic functions exhibit intrinsically local behavior: the implicit function theorem, analytic continuation, and branching phenomena are governed by local data. This suggests studying definability from a local perspective. Rather than asking whether a function is globally expressible in a given language, one

may ask whether its behavior on sufficiently small rational neighborhoods is controlled by a fixed family $\mathcal{F}$ of holomorphic functions.

In this report, we present Wilkie's work [4] on locally definable holomorphic functions. The guiding question is the following: given a set $\mathcal{F}$ of holomorphic functions, which holomorphic functions are locally definable from $\mathcal{F}$ in first-order logic? More precisely, to what extent does local definability determine a notion of analytic dependence, and what geometric structure does the associated closure operator induce?

Wilkie's investigation is also motivated by questions surrounding Zilber's program on complex exponentiation. Zilber [5] conjectured that the structure $(\mathbf{C}, +, \cdot, \exp)$ is quasiminimal, in the sense that every definable subset of $\mathbf{C}$ is either countable or co-countable. This conjectural picture connects model-theoretic tameness with deep transcendence phenomena such as Schanuel's conjecture (see Bays and Kirby [6]).

In the real case, Wilkie's model completeness and o-minimality results [3] for expansions of the real field by restricted analytic functions and the exponential function provide strong geometric control over definable sets. The idea suggested in his work is that one might attempt to study the unrestricted complex exponential function via its restrictions, which live inside an o-minimal structure. Since o-minimal structures admit a well-developed theory of complex analytic geometry (see Peterzil and Starchenko [7]), this approach offers a possible route toward understanding definability in the full complex exponential field.

The theory of locally definable holomorphic functions provides the necessary control over such restrictions and the analytic dependence they generate. In this sense, it supplies tools that may be relevant to the model-theoretic analysis of complex exponentiation.

2 First properties

In this section, we introduce locally definable functions and their properties. By definable, we mean definable by a first-order logic formula inside the language $\{0, 1, +, \times, <\}$ denoted by $\mathcal{L}$.

Definition 2.1 (Gaussian rational). The set of Gaussian rationals is $\mathbf{Q} + i\mathbf{Q}$.

Definition 2.2. Let f be a holomorphic function defined on an open subset U of $\mathbf{C}^n$. Let Δ be an open box with rational data, i.e., $\Delta = D_1 \times \cdots \times D_n$, where the D_i 's are open rectangles of $\mathbf{C}$ with Gaussian rational corners. We say that Δ is *suitable* for f (around $\underline{z}$) if $\overline{\Delta} \subset U$ (and $\underline{z} \in \Delta$) and in that case, $f|_{\Delta}$ is called a proper restriction of f .

Remark. Open boxes with rational data are the natural neighborhoods to consider from a logical point of view since they are definable within $(\mathbf{R}, \mathcal{L})$ (when we identify $\mathbf{C}$ with $\mathbf{R}^2$). Indeed, since $\mathcal{L}$ contains $\{0, 1, +, \times\}$, we can define any rational number (without parameters). Then, using the relation $<$, any interval with rational bounds can be defined. Therefore, any rectangle with Gaussian rational corners can be defined, and so can open boxes with rational data.

Let us fix a set $\mathcal{F}$ of holomorphic functions that contains polynomials with Gaussian rational coefficients in any number of variables. We denote by $\text{PR}(\mathcal{F})$ the set of all proper restrictions of elements of $\mathcal{F}$.

Definition 2.3. We denote by $\mathbf{R}(\text{PR}(\mathcal{F}))$ the extension of the ordered real line $(\mathbf{R}, \mathcal{L})$ with the graphs of functions of $\text{PR}(\mathcal{F})$. In other words, for any function $f: U \rightarrow \mathbf{C}$ with $U \in \mathbf{C}^n$ from $\text{PR}(\mathcal{F})$, we add a $(2n + 2)$ -relation $\mathcal{G}_f$ such that

$$(x_1, y_1, \dots, x_n, y_n, w_n, z_n) \in \mathcal{G}_f \iff w_n + iz_n = f(x_1 + iy_1, \dots, x_n + iy_n).$$

Remark. The structure $\mathbf{R}(\text{PR}(\mathcal{F}))$ is model-complete and $\mathcal{o}$ -minimal. Indeed, it is a reduct of $\mathbf{R}_{\text{an}}$, the structure containing $\mathcal{L}$ and real analytic functions. Then, Gabrielov's theorem [8] can be applied to $\mathbf{R}(\text{PR}(\mathcal{F}))$ and implies its model-completeness. Besides, a reduct of an $\mathcal{o}$ -minimal structure is $\mathcal{o}$ -minimal. This is part of the folklore.

Notice that restrictions to suitable neighborhoods is necessary to keep $\mathcal{o}$ -minimality. For instance, if $\mathcal{F}$ contains the exponential, then, adding the unrestricted exponential $\exp: \mathbf{C} \rightarrow \mathbf{C}$ to $(\mathbf{R}, \mathcal{L})$ leads to a structure not $\mathcal{o}$ -minimal. Indeed, we could define $\mathbf{Z}$ as the converse image of 1 by $\exp(i2\pi \cdot)$.

Definition 2.4. A holomorphic function f is *locally definable* from $\mathcal{F}$ if any proper restriction of f is definable in $\mathbf{R}(\text{PR}(\mathcal{F}))$.

Example 2.5. Suppose $\mathcal{F}$ contains $\exp$ and π . Then, any proper restriction of $\ln$ (the principal logarithm) is locally definable from $\text{PR}(\mathcal{F})$ using the following formula:

$$y = \ln(x) \text{ if and only if } x = \exp(y) \wedge |\Im(y)| < \pi.$$

Now, let us show that locally definable functions are stable under a couple of analytical operations.

Proposition 2.6. *If f is locally definable from $\mathcal{F}$, so are its partial derivatives.*

Proof. This follows from the fact that limits are definable. Indeed, let f be a function locally definable from $\mathcal{F}$. Then the $\partial_{z_1} f(\underline{z})$ is defined by the relation $\mathcal{G}$ (recall that functions are defined by their graphs as relations) such that $(\underline{x}, y) \in \mathcal{G}$ if and only if:

$$\forall \varepsilon > 0, \exists \eta > 0, \forall |h| < \eta, \left| y - \frac{f(\underline{x} + hz_1) - f(\underline{x})}{z_1} \right| < \varepsilon,$$

where f is definable by hypothesis. □

Proposition 2.7. *If f is locally definable from $\mathcal{F}$, so is its Schwarz reflection, i.e., the function $\underline{z} \mapsto f(\overline{\underline{z}})$.*

Proof. Let f be locally definable from $\mathcal{F}$. Then we can define its Schwarz reflection as the relation containing $(x_1, y_1, \dots, x_n, y_n, w_n, z_n)$ if and only if $(y_1, x_1, \dots, y_n, x_n, z_n, w_n)$ belongs to the graph of f . □

Definition 2.8. Let $f: U \rightarrow \mathbf{C}$ and $g: V \rightarrow \mathbf{C}$ be holomorphic functions such that $U \in \mathbf{C}^n$ and $V \in \mathbf{C}^{n+1}$. We say that f is implicitly defined from g if for any $\underline{z} \in V$, $(\underline{z}, f(\underline{z})) \in U$, $g(\underline{z}, f(\underline{z})) = 0$ and $\partial_{z_{n+1}} g(\underline{z}, f(\underline{z})) \neq 0$.

Proposition 2.9. *If f is implicitly defined from a function g locally definable from $\mathcal{F}$, then f is locally definable from $\mathcal{F}$.*

Proof. Let $f: U \rightarrow \mathbf{C}$ be a holomorphic function implicitly defined from a function $g: V \rightarrow \mathbf{C}$ locally definable from $\mathcal{F}$. Let $\underline{z} \in U$. By the implicit function theorem, there exists a neighborhood Δ suitable for f around $\underline{z}$ and a rectangle D such that $\Delta \times D$ is suitable for g around $(\underline{z}, f(\underline{z}))$ with the following property: for any $\underline{u} \in \Delta$, there exists a unique $u \in D$ such that $g(\underline{u}, u) = 0$. Since this u is equal to $f(\underline{z})$, we deduce that $f|_{\Delta}$ is definable in $\mathbf{R}(\text{PR}(\mathcal{F}))$ since g is definable in $\mathbf{R}(\text{PR}(\mathcal{F}))$. $\square$

Proposition 2.10. *If $f: U \rightarrow \mathbf{C}$, $g_1: V_1 \rightarrow \mathbf{C}$, ... and $g_n: V_n \rightarrow \mathbf{C}$ are locally definable from $\mathcal{F}$ where $U \subset \mathbf{C}^n$, then their composition $f \circ (g_1, \dots, g_n)$ (defined on $(g_1, \dots, g_n)^{-1}(U) \cap \bigcap_{i=1}^n V_i$) is locally definable from $\mathcal{F}$.*

Proof. Let $f, g_1, \dots$ and g_n be as above. Let us write W the domain of the function $f \circ (g_1, \dots, g_n)$. Let $\underline{z} \in W$. Let us consider an open box Δ suitable for f around $(g_1, \dots, g_n)(\underline{z})$. By continuity, the reciprocal image of Δ by $(g_1, \dots, g_n)$ is open and therefore contains an open box Δ' suitable for $(g_1, \dots, g_n)$ around $\underline{z}$. Hence $f \circ (g_1, \dots, g_n)|_{\Delta'}$ is definable in $\mathbf{R}(\text{PR}(\mathcal{F}))$. $\square$

Example 2.11. • If $\mathcal{F}$ is reduced to the set of polynomials with Gaussian rational coefficients, then rational fractions (with Gaussian rational coefficients) and algebraic functions (over rational fractions) are locally definable from $\mathcal{F}$.

- Any local inverse of a locally definable function from $\mathcal{F}$ is locally definable from $\mathcal{F}$.
- If $\mathcal{F}$ contains $\exp$ then functions such as $\ln$, $\cos$, $\sin$ and the Lambert function W (implicitly defined by $(w, z) \mapsto we^w - z$) are locally definable from $\mathcal{F}$. Moreover, so are solutions of classical functional equations such as $f + \exp(f) = z$ or the Kepler equation $f - e \sin(f) = M$ (where e and M are complex numbers). Elementary functions — functions obtained by iterating compositions of exponential, logarithm and algebraic functions — are also locally definable from $\mathcal{F}$ (see the work of Jaoui and Kirby [9]).

3 Pregeometries

In this section, we suppose $\mathcal{F}$ closed under differentiation and Schwartz reflection. We introduce pregeometry operators naturally associated to locally definable functions.

Definition 3.1. We denote by $\tilde{\mathcal{F}}$ the closure of $\mathcal{F}$ under composition and implicit definability.

In particular, any function from $\tilde{\mathcal{F}}$ is locally definable from $\mathcal{F}$ according to Propositions 2.9 and 2.10.

Definition 3.2. Let X be a subset of $\mathbf{C}$. We denote by $\tilde{D}(X)$ the set of complex numbers of the form $f(\underline{z})$, where $f \in \tilde{\mathcal{F}}$ and $\underline{z} \in X^*$ belongs to the domain of f .

Definition 3.3. Let X be a subset of $\mathbf{C}$. We denote by $LD(X)$ the set of complex numbers of the form $f(\underline{z})$, where f is locally definable from $\mathcal{F}$ and $\underline{z} \in X^*$ belongs to the domain of f .

Theorem 3.4. *The operators $\tilde{D}$ and LD are pregeometries.*

Remark. Theorem 3.4 highlights the analogy between the algebraic closure and the operators $\tilde{D}$ and LD . Indeed, while the algebraic closure of $X \subset \mathbf{C}$ is obtained by solving polynomial equations with coefficients in X , we can think about $\tilde{D}(X)$ and $LD(X)$ obtained by solving some functional analytic equations with coefficients in X .

Lemma 3.5. *For any open subset $U \subset \mathbf{C}^*$, $\iota: U \rightarrow \mathbf{C}, z \mapsto 1/z$ belongs to $\tilde{\mathcal{F}}$. The set $\tilde{\mathcal{F}}$ is closed under differentiation.*

Proof. First of all, let us fix U as above. Let us consider the polynomial $p: (x, y) \mapsto xy - 1$ which belongs to $\tilde{\mathcal{F}}$ since it has Gaussian rational coefficients. We have $p(z, 1/z) = 0$ and $\partial_y p(z, 1/z) = z \neq 0$ when $z \in \mathbf{C}^*$. Hence, ι is implicitly defined from p and belongs to $\tilde{\mathcal{F}}$.

Let $\mathcal{S}$ be the set of functions f of $\tilde{\mathcal{F}}$ whose first-order partial derivatives belong to $\tilde{\mathcal{F}}$. Remember that we assumed $\mathcal{F}$ to be closed under differentiation. Hence, we have $\mathcal{F} \subset \mathcal{S} \subset \tilde{\mathcal{F}}$ and it is enough to show that $\mathcal{S}$ is closed under implicit definability and composition.

First, let $f: U \rightarrow \mathbf{C}$ be a holomorphic function implicitly defined from a function $g: V \rightarrow \mathbf{C}$ belonging to $\mathcal{S}$. In particular, differentiating $g(\underline{z}, f(\underline{z}))$, we get for any $i \in \{1, \dots, n\}$:

$$\partial_{z_i} g(\underline{z}, f(\underline{z})) + \partial_{z_i} f(\underline{z}) \partial_{z_{n+1}} g(\underline{z}, f(\underline{z})) = 0,$$

so

$$\partial_{z_i} f(\underline{z}) = -\partial_{z_i} g(\underline{z}, f(\underline{z})) \partial_{z_{n+1}} g(\underline{z}, f(\underline{z}))^{-1}.$$

Since we showed that ι belongs to $\tilde{\mathcal{F}}$, by closure under composition, we deduce that first-order partial derivatives of f belong to $\tilde{\mathcal{F}}$. In other words, f belongs to $\mathcal{S}$.

Let $f: U \rightarrow \mathbf{C}$, $g_1: V_1 \rightarrow \mathbf{C}$, ... and $g_n: V_n \rightarrow \mathbf{C}$ be functions of $\mathcal{S}$, with $U \subset \mathbf{C}^n$ and $V_i \subset \mathbf{C}^{n_i}$ for all $i \in \{1, \dots, n\}$. We denote by W the domain of $f \circ (g_1, \dots, g_n)$. We have for any $(\underline{z}, \underline{w}) \in W$ with $\underline{z} \in \mathbf{C}^{n_1}$ and for any $i \in \{1, \dots, n_1\}$:

$$\partial_{z_i} f \circ (g_1, \dots, g_n)(\underline{z}, \underline{w}) = \partial_{z_i} g_1(\underline{z}) \cdot (\partial_{z_1} f) \circ (g_1, \dots, g_n)(\underline{z}, \underline{w}),$$

where $\partial_{z_i} g_1$, $\partial_{z_1} f$, g_1 , ... and g_n belong to $\tilde{\mathcal{F}}$ by definition of $\mathcal{S}$. Hence, since $\tilde{\mathcal{F}}$ is closed under composition, the right-hand term belongs to $\tilde{\mathcal{F}}$. This reasoning applies to $i > n_1$ and therefore the composite function belongs to $\mathcal{S}$. $\square$

proof of Theorem 3.4. Let X be a subset of $\mathbf{C}$.

It is clear that $X \subset \text{LD}(X)$ since the identity belongs to $\mathcal{F}$ (it is a polynomial with Gaussian coefficients).

Any element of $\text{LD}(X)$ is of the form $f(\underline{z})$ where f is locally definable from $\mathcal{F}$ and $\underline{z} \in X^*$ and therefore belongs to $\text{LD}(\underline{z})$. In other words, $\text{LD}(X)$ is the union of $\text{LD}(\underline{z})$ where $\underline{z} \in X^*$.

It follows from Proposition 2.10 that $\text{LD}(\text{LD}(X)) = \text{LD}(X)$.

Let a and b be complex numbers such that $a \in \text{LD}(X \cup \{b\}) \setminus \text{LD}(X)$. By Definition 3.3, there exists a function f locally definable from $\mathcal{F}$ such that $a = f(\underline{w}, b)$ and $\underline{w} \in X^n$. Suppose that $f(\underline{w}, \cdot)$ is constant on a neighborhood of b (with value a). Let q be a Gaussian rational in this neighborhood. Then, the function $\underline{z} \mapsto f(\underline{z}, q)$ is locally definable from $\mathcal{F}$ and takes the value a in $\underline{w}$. However, a does not belong to $\text{LD}(X)$. By contradiction, there exists $i \in \mathbf{N}^*$ minimal such that $\partial_{z_{n+1}}^i f$ is non-zero on $(\underline{w}, b)$. We can suppose $i = 1$ without loss of generality because Proposition 2.10 and Proposition 2.6 allow us to replace f by $f + \partial_{z_{n+1}}^{i-1} f$ when $i > 1$ (because this function equals a in $(\underline{w}, b)$ since the second term vanishes). Now, let $g: (\underline{z}, z_{n+1}, z_{n+2}) \mapsto f(\underline{z}, z_{n+2}) - z_{n+1}$ be a function locally definable from $\mathcal{F}$. We have $g(\underline{w}, a, b) = 0$ and $\partial_{z_{n+2}} g(\underline{w}, a, b) \neq 0$. Consequently, we can apply the implicit function theorem and find a holomorphic function h implicitly defined from g such that $h(\underline{w}, a) = b$. In particular, h is locally definable from $\mathcal{F}$ according to Proposition 2.9 and therefore $b \in \text{LD}(X \cup \{a\})$.

Consequently, LD is a pregeometry. The same line of reasoning applies to $\tilde{\text{D}}$ except that one needs to invoke Lemma 3.5 instead of Proposition 2.6 in the proof above. □

Lemma 3.6. *Let $\underline{w} = (w_1, \dots, w_n) \in \mathbf{C}^n$. Then, $w_1, \dots$ and w_n are $\tilde{\text{D}}$ -dependent (respectively LD -dependent) if and only if there exists $f \in \tilde{\mathcal{F}}$ (respectively a function f locally definable from $\mathcal{F}$) such that $\underline{w}$ belongs to the domain of f , $f(\underline{w}) = 0$ and f does not vanish identically on any neighborhood of $\underline{w}$.*

Proof. Let $\underline{w} = (w_1, \dots, w_n) \in \mathbf{C}^n$.

Suppose $w_1 \in \text{LD}(w_2, \dots, w_n)$. Then, by Definition 3.3, there exists a function f locally definable from $\mathcal{F}$ such that $w_1 = f(w_2, \dots, w_n)$. The function $\underline{z} \mapsto f(z_2, \dots, z_n) - z_1$ is locally definable from $\mathcal{F}$, vanishes at $\underline{w}$ and does its first-order derivative relatively to its first variable is non-zero in $\underline{w}$.

Conversely, suppose there is a function f locally definable from $\mathcal{F}$ such that $f(\underline{w}) = 0$ and f does not vanish identically on any neighborhood of $\underline{w}$. Without loss of generality, we can suppose $\partial_{z_1} f(\underline{w}) \neq 0$ (as we did for the proof of Theorem 3.4, f can be replaced by another function if needed). Now, by the implicit function theorem, we can find a function g implicitly defined from f (and so locally definable from $\mathcal{F}$) such that $g(w_2, \dots, w_n) = w_1$. So $w_1 \in \text{LD}(w_2, \dots, w_n)$.

The same line of reasoning applies to $\tilde{\text{D}}$. □

4 Derivations

In this section, $\mathbf{K}$ denotes either $\mathbf{R}$ or $\mathbf{C}$. We fix $\mathcal{C}$ a set of real analytic functions when $\mathbf{K} = \mathbf{R}$ or holomorphic when $\mathbf{K} = \mathbf{C}$. We introduce derivations and give some of their properties.

Definition 4.1 (Derivation). A derivation is a $\mathbf{Q}$ -linear map which is also $\mathbf{Q}(i)$ -linear when $\mathbf{K} = \mathbf{C}$.

Let f be an analytic function on an open subset of $\mathbf{K}^n$ and $\underline{a} = (a_1, \dots, a_n)$, a point of the domain of f . A derivation δ respects f at $\underline{a}$ if

$$\delta(f(a)) = \sum_{i=1}^n \partial_i f(a) \cdot \delta(a_i).$$

A derivation δ respects f if it respects f at all points of its domain.

We denote by $\text{Der}_{\mathbf{K}}(\mathcal{C})$ the set of derivations that respect all $f \in \mathcal{C}$.

Remark. A derivation in the usual sense, i.e., that satisfies Leibniz rule, is a derivation in the sense of Definition 4.1 that respects multiplication. Indeed, let δ be such a derivation and let $\mu: \mathbf{C}^2 \rightarrow \mathbf{C}$ be the multiplication map. Then, $\partial_1 \mu(x, y) = y$ and $\partial_2 \mu(x, y) = x$ so we get $\delta(x \times y) = y\delta(x) + x\delta(y)$ for any $(x, y) \in \mathbf{C}^2$.

Definition 4.2. For a finite subset X of $\mathbf{K}$ let

$$\text{DD}_{\mathbf{K}}^{\mathcal{C}}(X) = \{a \in \mathbf{K}, \forall \delta \in \text{Der}_{\mathbf{K}}(\mathcal{C}), \delta(X) = \{0\} \implies \delta(a) = 0\}.$$

For an arbitrary subset X of $\mathbf{K}$, we extend the definition by

$$\text{DD}_{\mathbf{K}}^{\mathcal{C}} = \bigcup_{Y \subset X, |Y| < \infty} \text{DD}_{\mathbf{K}}^{\mathcal{C}}(Y).$$

Remark. In Definition 4.2, $\text{DD}_{\mathbf{K}}^{\mathcal{C}}$ on arbitrary sets does extend $\text{DD}_{\mathbf{K}}^{\mathcal{C}}$ on finite sets. This follows from the fact that $\text{DD}_{\mathbf{K}}^{\mathcal{C}}$ (on finite sets) is non-decreasing from $\mathcal{P}(\mathbf{K})$ to $\mathcal{P}(K)$. Indeed, let $Y \subset X \subset \mathbf{K}$, $a \in Y$ and $\delta \in \text{Der}_{\mathbf{K}}(\mathcal{C})$ such that $\delta(X) = \{0\}$. Then, in particular, $\delta(Y) = \{0\}$. Therefore, $\delta(a) = 0$. So $a \in X$.

Lemma 4.3. The operator $\text{DD}_{\mathbf{K}}^{\mathcal{C}}$ is a pregeometry.

Proof. Let X be a subset of $\mathbf{K}$.

It is clear that $X \subset \text{DD}_{\mathbf{K}}^{\mathcal{C}}(X)$.

The axiom of finite character is part of the definition.

Let $a \in \text{DD}_{\mathbf{K}}^{\mathcal{C}}(\text{DD}_{\mathbf{K}}^{\mathcal{C}}(X))$ and let $\delta \in \text{Der}_{\mathbf{K}}(\mathcal{C})$ such that $\delta(X) = \{0\}$. By definition, $\delta(\text{DD}_{\mathbf{K}}^{\mathcal{C}}(X)) = \{0\}$ so in particular, $\delta(a) = 0$. Hence, $a \in \text{DD}_{\mathbf{K}}^{\mathcal{C}}(X)$ and we conclude that $\text{DD}_{\mathbf{K}}^{\mathcal{C}}(\text{DD}_{\mathbf{K}}^{\mathcal{C}}(X)) = \text{DD}_{\mathbf{K}}^{\mathcal{C}}(X)$ (the other inclusion comes from the first axiom).

Let a and b be elements of $\mathbf{K}$ such that $a \notin \text{DD}_{\mathbf{K}}^{\mathcal{C}}(X)$ and $b \notin \text{DD}_{\mathbf{K}}^{\mathcal{C}}(X \cup \{a\})$. We can suppose without loss of generality that X is finite. By Definition 4.2,

there exists $(\delta_1, \delta_2) \in \text{Der}_{\mathbf{K}}(\mathcal{C})^2$ such that $\delta_1(X) = \delta_2(X \cup \{a\}) = \{0\}$, $\delta_1(a) \neq 0$ and $\delta_2(b) \neq 0$. Let $\delta = \delta_2(b)\delta_1 - \delta_1(b)\delta_2$ be an element of $\text{Der}_{\mathbf{K}}(\mathcal{C})$ (which is a $\mathbf{K}$ -vector space). Then, since δ_1 and δ_2 vanish on X , so does δ . Moreover, we have $\delta(b) = 0$. In other words, we have $\delta(X \cup \{b\}) = \{0\}$. However, we have $\delta(a) = \delta_2(b)\delta_1(a) \neq 0$. Therefore, $a \notin \text{DD}_{\mathbf{K}}^{\mathcal{C}}(X \cup \{b\})$. $\square$

Now, we give some results without proof (see Wilkie [4] for proofs).

Theorem 4.4. *The pregeometries $\text{DD}_{\mathbf{C}}^{\mathcal{F}}$ and $\tilde{\mathbf{D}}$ coincide.*

Lemma 4.5. *We have $\text{Der}_{\mathbf{K}}(\mathcal{C}) = \text{Der}_{\mathbf{K}}(\tilde{\mathcal{C}})$, where $\tilde{\mathcal{C}}$ denotes functions locally definable from $\mathcal{C}$.*

Definition 4.6. Let λ and μ be functions from $\mathbf{R}$ to $\mathbf{R}$. We define

$$[\lambda : \mu]: \begin{array}{ccc} \mathbf{C} & \longrightarrow & \mathbf{C} \\ x + iy & \longmapsto & (\lambda(x) - \mu(y)) + i(\lambda(y) + \mu(x)) \end{array} .$$

Theorem 4.7. *Suppose $\mathcal{C}$ is a collection of holomorphic functions closed under Schwartz reflection. Then we have:*

$$\text{Der}_{\mathbf{C}}(\mathcal{C}) = \{[\lambda : \mu], (\lambda, \mu) \in \text{Der}_{\mathbf{R}}(\Re(\mathcal{C}) \cup \Im(\mathcal{C}))\}.$$

Remark. The above theorem can be intuited thanks to the Cauchy-Riemann equations. Indeed, when λ and μ are real derivations, the real matrix of the linear map $[\lambda : \mu]$ belongs to $\mathbf{R}^+ \text{SO}_2(\mathbf{R})$

5 Conjecture

In this section, we introduce the main conjecture of Wilkie [4].

Conjecture 5.1. *A function f is locally definable from $\mathcal{F}$ if and only if for any $\underline{z}$ in the domain of f there exist an open box Δ suitable for f around $\underline{z}$ and a function $g \in \tilde{\mathcal{F}}$ such that $f|_{\Delta} = g|_{\Delta}$.*

The link between real and complex derivations drawn in Section 4 can be used to deduce the following theorem:

Theorem 5.2. *The pregeometries $\tilde{\mathbf{D}}$ and LD coincide. Furthermore, Conjecture 5.1 holds in neighborhoods of generic points, i.e., if $z_1, \dots$ and z_n are independent complex numbers, and f is a function locally definable from $\mathcal{F}$ whose domain contains $\underline{z} = (z_1, \dots, z_n)$, then there exist $g \in \tilde{\mathcal{F}}$ and an open box Δ both suitable for f and g around $\underline{z}$ such that $f|_{\Delta} = g|_{\Delta}$.*

Sketch proof. The work above can be done for real analytic functions, introducing pregeometries $\tilde{\mathbf{E}}$ and LED analogous to $\tilde{\mathbf{D}}$ and LD by replacing "holomorphic" by "analytic" in Definition 3.2 and Definition 3.3 respectively. Hence, we

first claim (see Wilkie [4] for proof) that $\tilde{E}$ and LED coincide. In the real analytic case, Gabrielov's theorem [8] implies that the structure $\mathbf{R}(\text{PR}(\mathcal{E}))$ (analogous to $\mathbf{R}(\text{PR}(\mathcal{F}))$ where $\mathcal{E}$ denotes a set of real analytic functions containing polynomials with rational coefficients) is model-complete. Then, we claim (see Wilkie [4] for proof) that $\text{PR}(\mathcal{E})$ is existentially closed in $\mathbf{R}(\text{PR}(\mathcal{E}))$. Thus, by model-completeness, this implies that $\text{PR}(\mathcal{E})$ is an elementary substructure of $\mathbf{R}(\text{PR}(\mathcal{E}))$. In particular, $\text{PR}(\mathcal{E})$ is closed under $\mathbf{R}(\text{PR}(\mathcal{E}))$ -definable functions. It follows from $\tilde{E} = \text{LED}$ and the real analogous of Lemma 3.6 that for any $\tilde{E}$ -independent set $\{s_1, \dots, s_n\}$ any function g locally definable from $\mathcal{E}$ whose domain contains $\underline{s} = (s_1, \dots, s_n)$ that there exists a function $g \in \tilde{E}$ that coincides with f on a neighborhood of $\underline{s}$. Then, one can prove that any $\lambda \in \text{Der}_{\mathbf{R}}(\mathcal{E})$ respects all functions locally definable from $\mathcal{E}$, using Lemma 4.5 on generic points.

Now, we can complete the proof. Let $X \subset \mathbf{C}$. Thanks to Theorem 4.4, it suffices to show $\text{LD}(X) \subset \text{DD}(X)$. So, let $w \in \text{LD}(X)$, f be a function locally definable from $\mathcal{F}$ such that $w = f(\underline{z})$ where $\underline{z} = (z_1, \dots, z_n)$ is a tuple of elements of X . Let δ be a derivation of $\text{Der}_{\mathbf{C}}(\mathcal{F})$ vanishing on $\underline{z}$. By Theorem 4.7, if we let $\mathcal{E}$ be the set $\mathcal{R}(\mathcal{F}) \cup \mathcal{S}(\mathcal{F}) \cup \mathbf{Q}[X_0, X_1, \dots]$, δ can be written as $\delta = [\lambda : \mu]$ where λ and μ belong to $\text{Der}_{\mathbf{R}}(\mathcal{E})$. Now, $\mathcal{R}(f)$ and $\mathcal{S}(f)$ are locally definable from $\mathcal{E}$ and therefore respected by λ and μ by the discussion above. In particular, it can be shown that this implies that δ respects f . $\square$

Example 5.3. When $\mathcal{F}$ is the set of polynomials with Gaussian rational coefficients, then the conjecture holds on neighborhoods of points $(z_1, \dots, z_n)$ such that $z_1, \dots$ and z_n are algebraically independent over $\mathbf{Q}$.

6 Conclusion

The theory of locally definable holomorphic functions reveals an underlying geometry of analytic dependence generated by a fixed family $\mathcal{F}$. The locally definable closure operator satisfies the axioms of a pregeometry, thereby allowing one to define dimension and independence in a manner formally analogous to algebraic dependence in fields.

A central insight of Wilkie's approach is that this geometric structure admits an infinitesimal characterization via derivations. An element belongs to the closure of a set precisely when it admits no independent infinitesimal variation compatible with the analytic relations imposed by $\mathcal{F}$. In this way, analytic dependence is detected through rigidity under all $\mathcal{F}$ -derivations.

The resulting framework places local definability within a broader geometric perspective: definability is not merely a syntactic notion, but gives rise to a structured dependence relation governed by analytic constraints. Understanding the extent to which locally definable functions coincide generically with explicitly constructible analytic functions remains a central problem in this area.

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